```
-- case when statements
-- if-elseif-else


syntax:


case
  when condition1 then result1
  when condition2 then result2
  when condition3 then result3
  else default_result
end


select *
from `e1.customers`
limit 5


categorise customers as
18-20: genz,
21-40 : adult,
41 onwards : senior_citizen
```

```sql
select
  CustomerID,
  Age,
  Name,
  case
    when Age >= 18 and Age <= 20 then "GenZ"
    when Age >= 21 and Age <= 40 then "Adult"
    else "Senior_Citizen"
  end as age_group
from `e1.customers` as c
```

```sql
select
  CustomerID,
  Age,
  Name,
  case
    when Age >= 18 and Age <= 20 then "GenZ"
    when Age <= 40 then "Adult"
    when Age >40 then "Senior_Citizen"
  end as age_group
from `e1.customers` as c
```

```sql
select min(age) from `e1.customers`


select *
from `e1.customers`
limit 5


F --> Female
M --> Male

select
    CustomerID,
    Name,
    Gender,
    case
        when Gender = 'M' then 'Male'
        else 'Female'
    end as Gender_New
from `e1.customers`


For every order, calculate the total order value by adding the
Total column from order_items.


Then classify each order as:
```

```
High Value → Total order value ≥ ₹50,000
Medium Value → Total order value between ₹20,000 and ₹49,999
Low Value → Total order value < ₹20,000


Display:


OrderID --> orders
OrderDate --> orders
CustomerName --> customer
TotalOrderValue --> order_items
OrderCategory --> create using total_sales


Show the highest-value orders first. --> order by



-- cust, order --> custid
-- order, order_item --> orderid

select
  o.OrderID,
  o.OrderDate,
  c.Name,
  sum(oi.Total) as total_sales,
  case
    when sum(oi.Total) >= 50000 then "High Value"
    when sum(oi.Total) >= 20000 then "Medium Value"
```

```
    else "Low Value"
  end as OrderCategory
from `e1.orders` as o
inner join `e1.customers` as c
  on o.CustomerID = c.CustomerID
inner join `e1.order_items` as oi
  on o.OrderID = oi.OrderID
group by
  o.OrderID,
  o.OrderDate,
  c.Name
order by total_sales desc
```

Problem Statement

Create a category-level sales report containing:

CategoryID --> category
CategoryName --> category
Number of products in the category --> product
Number of different products sold --> order_items
Total quantity sold --> order_items
Total sales value --> order_items
Average selling price --> order_items

```
Only consider order-item records where:



SellingPrice > ₹10,000 --> where



Display only categories where:



Total sales value > ₹5,00,000 --> having



Sort categories from highest to lowest total sales. --> order by




select
  c.CategoryID,
  c.CategoryName,
  count(distinct p.ProductID) as total_products,
  count(distinct oi.ProductID) as different_products_sold,
  sum(oi.Quantity) as total_qty,
  sum(oi.Total) as total_sales,
  round(avg(oi.SellingPrice),2) as avg_selling_price
from `e1.categories` as c
inner join `e1.products` as p
on p.CategoryID = c.CategoryID
inner join `e1.order_items` as oi
on oi.ProductID = p.ProductID
where oi.SellingPrice > 1000
```

```
group by
   c.CategoryID,
  c.CategoryName
having total_sales > 5000
order by total_sales desc




Find products where:

MRP is greater than ₹15,000 --> where
Discount percentage is greater than 20% -->
Total sales value is greater than ₹75,000

Display:

ProductID --> product
ProductName --> product
CategoryName --> category
MRP --> product
SellingPrice --> product
Discount percentage --> derieved field --> (mrp - sp)*100/mrp
Total sales --> order_items

Sort by discount percentage from highest to lowest.



select
  p.ProductID,
```

```
  p.ProductName,
  c.CategoryName,
  p.MRP,
  p.SellingPrice,
  ((p.MRP - p.SellingPrice)*100)/p.MRP as discount_percentage,
  sum(oi.Total) as total_sales
from `e1.products` as p
inner join `e1.categories` as c
on p.CategoryID = c.CategoryID
inner join `e1.order_items` as oi
on p.ProductID = oi.ProductID
where p.MRP > 15000
group by
  p.ProductID,
  p.ProductName,
  c.CategoryName,
  p.MRP,
  p.SellingPrice
having
   ((p.MRP - p.SellingPrice)*100)/p.MRP > 20
   and total_sales > 75000
order by discount_percentage desc
```

For every employee, calculate:

EmployeeID

```
Employee name
Department
Number of orders handled
Number of orders whose total value exceeds ₹50,000
Total sales value of all orders handled

Display only employees who have handled at least 5 orders.

Sort employees by total sales value from highest to lowest.
```